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Introduction to Data Quality and Data Cleaning

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Agenda

- **Introduction**
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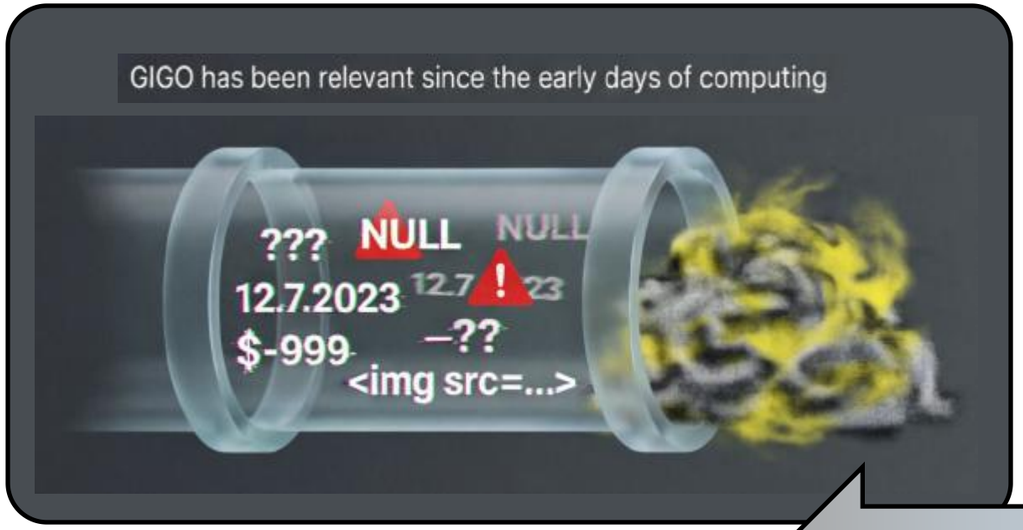
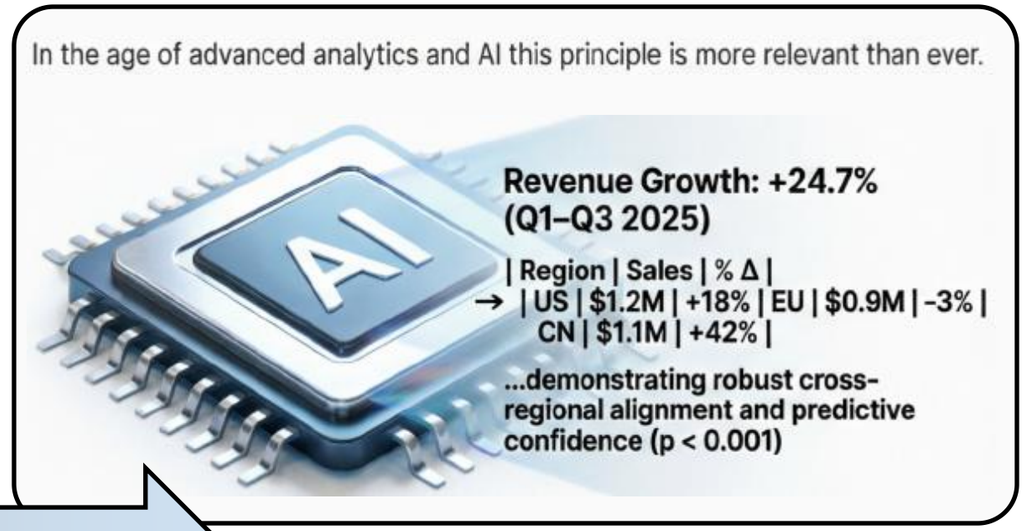
  - Challenges in extracting insights from Data
  - Real World Consequences of Poor Data Quality
- **Definition of Data Quality**
  - Definition
  - Data Quality Dimensions
- **Measuring Data Quality**
  - Measure intrinsic quality dimensions
- **Data Quality in the Age of AI**
  - Redefining Quality
- **Data Processes**
  - Data Wrangling, Cleaning, and Profiling

The Enduring Truth: Garbage In, Garbage Out

Input



Output



A convincing output do not guarantee high quality input

With powerful AI models the garbage is getting harder to spot, and extracting the right insights become more challenging.

Challenges in extracting insights from Data *(Especially for AI Success)*

- Large datasets **don't always guarantee** valuable insights
- Managing and processing Data can be **complex and resource-intensive**
- Existing analytics algorithms often **need adaptation** to handle large Data requirements
- Data often **lacks** predefined **schemas and quality properties**

80% of AI Effort Is Spent on Data

Data collection and preparation consume around 80 percent of total AI development effort

Oversimplified Metrics

Quality is often measured in oversimplified ways: sample counts, class balance, missing values

Real-World Mismatch

Training data may not reflect real world conditions leading to underperformance

Governance Gaps Accelerate Decay

Data governance and documentation are frequently overlooked causing quality degradation

Unreviewed Sources = Hidden Risks

Non reviewed data sources like web scraping carry biases, inaccuracies or misinformation

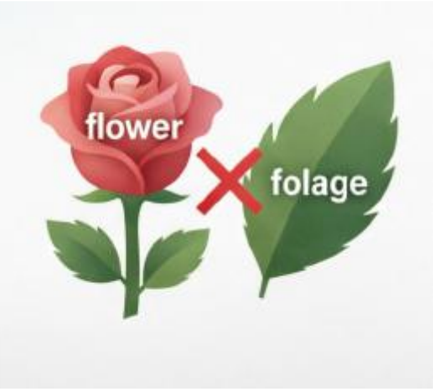
Real World Consequences of Poor Data Quality

Gender Bias in AI

AI models advised women to ask for lower salaries than men

Semantic Misunderstanding

Plant color recorded as green 85% of time but field meant flower color not foliage

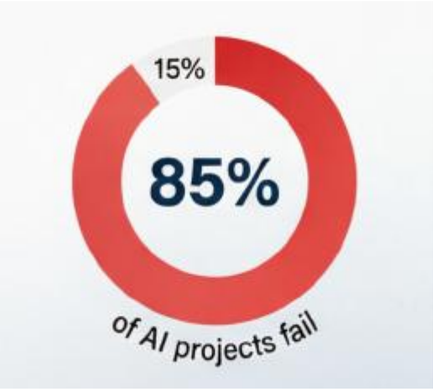


Timeliness Mismatch

Aerial photos recent but digital map labels several years outdated causing alignment issues

Gartner Estimate

Up to 85% of AI projects would fail, often due to poor data quality



Definition of Data Quality

- ***Data Quality (DQ)*** is a complex concept that varies across contexts, domains, and fields of application (academia, industry, …)

It is generally defined as:

1. The capability of **data to satisfy stated and implied needs** when used under specified conditions
2. The **fitness of data for its intended use** ("fitness for use")
3. A property related to **quality management**, where **data must meet user needs** and reflect the reality it intends to represent

- DQ is highly ***dependent on the source of the data***, as well as the **processes used to collect, process, and analyze it**.

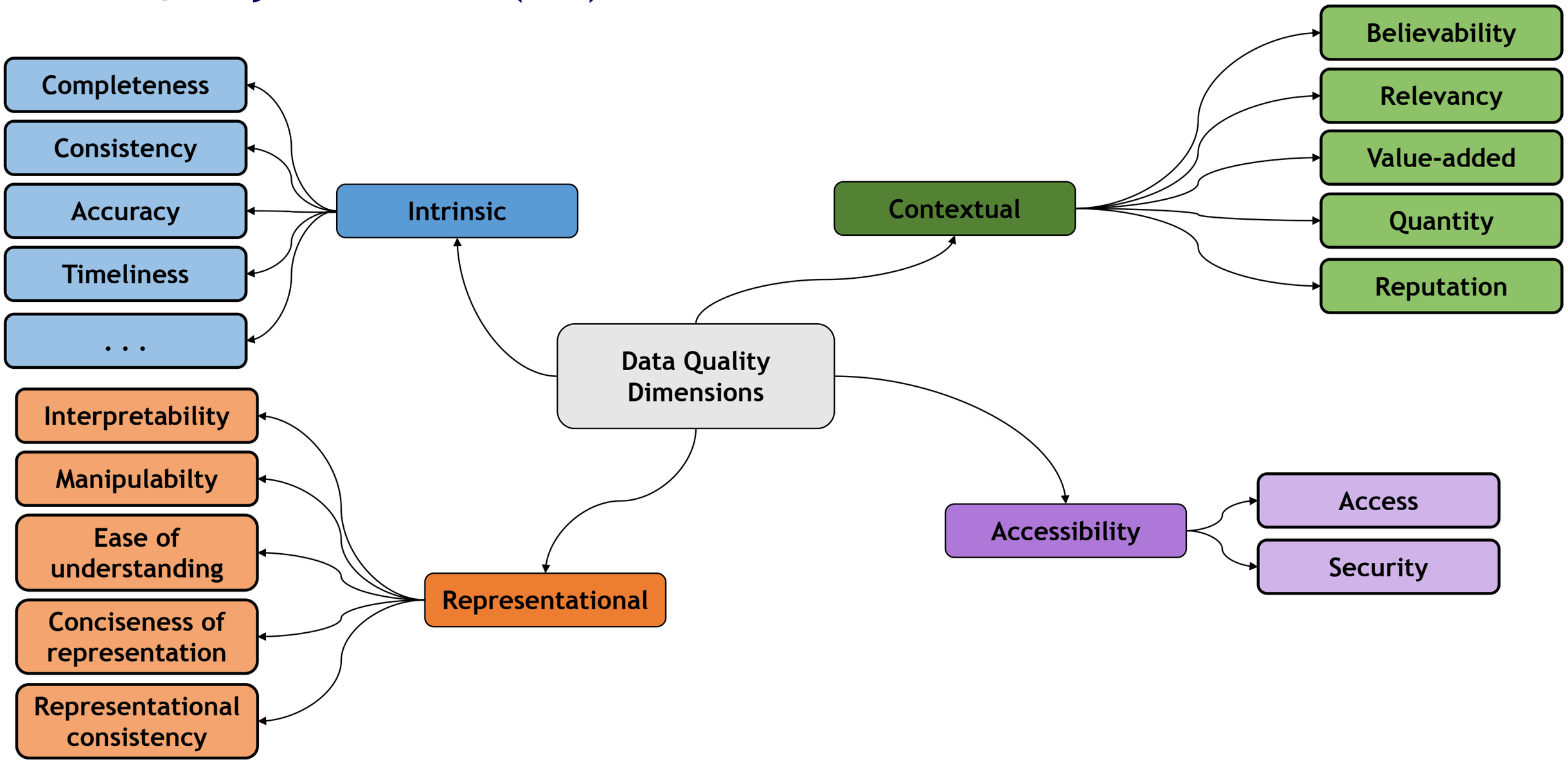
Data Quality Dimensions (1/3)

- DQD's are used to **measure**, **quantify**, and **manage DQ**

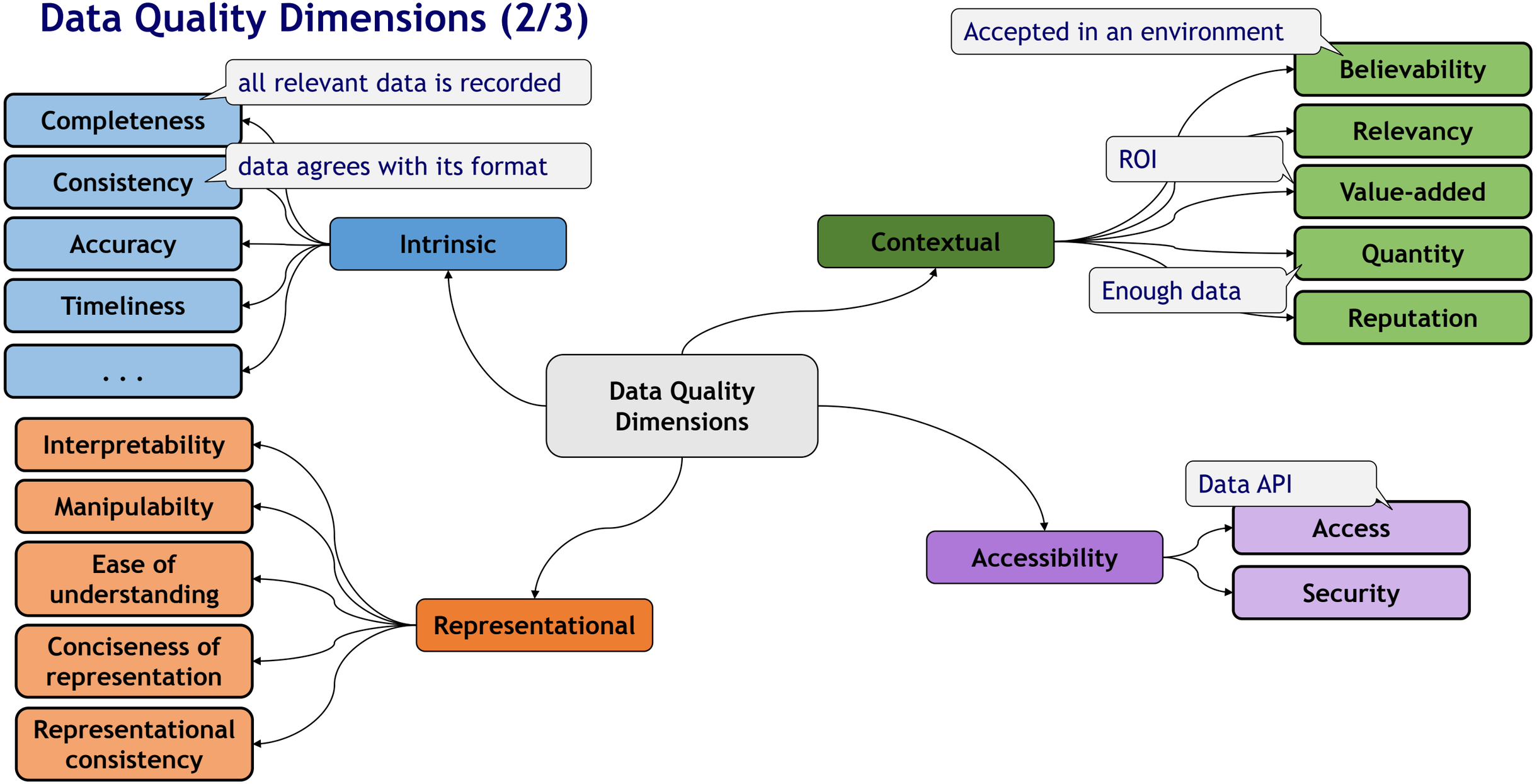
- Each **dimension** has a specific **metric**, which **measures its performance**

- DQDs are classified into **4 categories**:
  - **Intrinsic**: Refers to the inherent properties of the data itself
  - **Contextual**: Depends on the data's purpose and relevance to specific use cases
  - **Accessibility**: Concerns data availability, access, and security
  - **Representational**: How data is presented and understood

Data Quality Dimensions (2/3)



Data Quality Dimensions (2/3)



Data Quality Dimensions (3/3)

- Examples of intrinsic quality dimensions:

DQD's	Description
Completeness	• Describes whether all relevant data is recorded • Measures missing values for an attribute
Consistency	• Checks whether data agrees with its format and structure • Refers to the respect of data constraints
Accuracy	• Measures whether data was recorded correctly and reflects realistic values • Defined as the "closeness of the agreement between the result of a measurement and a true value of the measure"
Timeliness	• Computes whether data is up to date, referring to data currency and volatility
Uniqueness	• Every record represents a distinct entity or event

Measuring Accuracy and Completeness

Accuracy

How closely do data values match real world truth?

Metrics:
- Percentage of records matching trusted source
- Error rate in sampled records

Tools:
- Great Expectations, Datafold, Deequ, Monte Carlo

Completeness

Are all required data fields populated?

Formula:
- Non-null values ÷ Total expected values

Tools:
- Great Expectations, Talend, dbt

Measuring Consistency and Timeliness

Consistency

Is data uniform across different sources and formats?

Metric:
- % of values consistent across systems

Method:
- Compare same entities across multiple datasets
- Check conflicting entries

Timeliness

How current is data relative to expected refresh schedule?

Metric:
- Average data latency, time difference between event and record availability

Tools:
- Monte Carlo, Bigeye, Atlan

Measuring Uniqueness and Reliability

Uniqueness

Are there duplicate records representing the same entity?

Formula:
- Duplicate records ÷ Total records

Tools:
- OpenRefine, Great Expectations, Datafold

Reliability

How consistently is data accurate and available over time?

Validation Methods:
- Validate against regex patterns
- Validate against data type constraints
- Validate against domain rules

Measuring Usefulness and Differences

Usefulness

Does data support real business decisions and outcomes?

Tools:

- Atlan, Collibra, Alation, Datafold

Differences

Proportion of rows, columns or values that differ between two datasets

Method:

- Run dataset comparison using hashing or row-by-row checks

Approaches:

- Hash-Based Compare or Exact Row Match

Metrics & Measurements

Measuring data quality is essential once data is generated to ensure reliability

- **Data Quality Dimensions (DQD)** must be must be relevant to identified data quality problems and quantified to assess performance
- **Quality metric** is a quantitative or categorical representation of one or more attributes

  Quality metrics evaluate if attribute values align with a targeted quality dimension
- **Measurements** are performed for each attribute based on its type and data ranges

  Measurements often produces binary results (**correct/incorrect**) or a percentage scale (**0-100%**)
- **Common metrics are often used to evaluate dimensions such as accuracy, completeness, consistency, and currency of data**

DQD's	Metric functions
Completeness	Comp(%) = NNMV*100/N: Number of non-missing values /N
Consistency	Cons(%) = NVRC*100/N: Number of values that respects constraints /N
Accuracy	Acc(%) = NCV*100/N: Number of correct values /N
Uniqueness	Uniq(%) = NDV*100/N: Number of distinct values /N

*N: Total number of observations (Rows) in dataset or sample

- **Specific metrics** must be defined for non-structured data types (images, videos, audio)

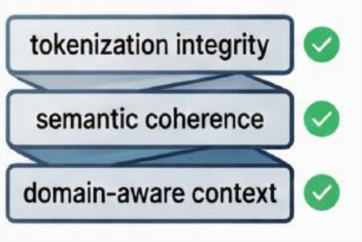
Modern AI is Redefining Quality

- Because the rise of Generative AI (GenAI), Large Language Models (LLMs), and autonomous agentic systems has changed how data is consumed.

Traditional DQ		AI-era DQ
human-readable reporting		machine-interpreted patterns
Accuracy & consistency correctness and uniformity	shift →	**Context & nuance** representation and semantics
Structured reports tabular output for decisions		**Machine patterns** model-ready, not report-ready
Periodic monitoring batch checks, reactive fixes		**Continuous monitoring** prevents hallucinations & bias

Modern AI is Redefining Quality

Text Data Requirements



Text data has different requirements for successful utilization.

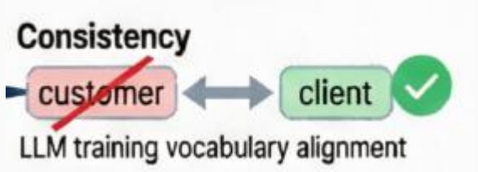
Machine-readable structure and contextual richness are now foundational.

Machine Actors Depend on Quality

Machine actors now directly rely on data being of good quality.

No human-in-the-loop fallback, quality failures propagate instantly.

Traditional Dimensions Recontextualized

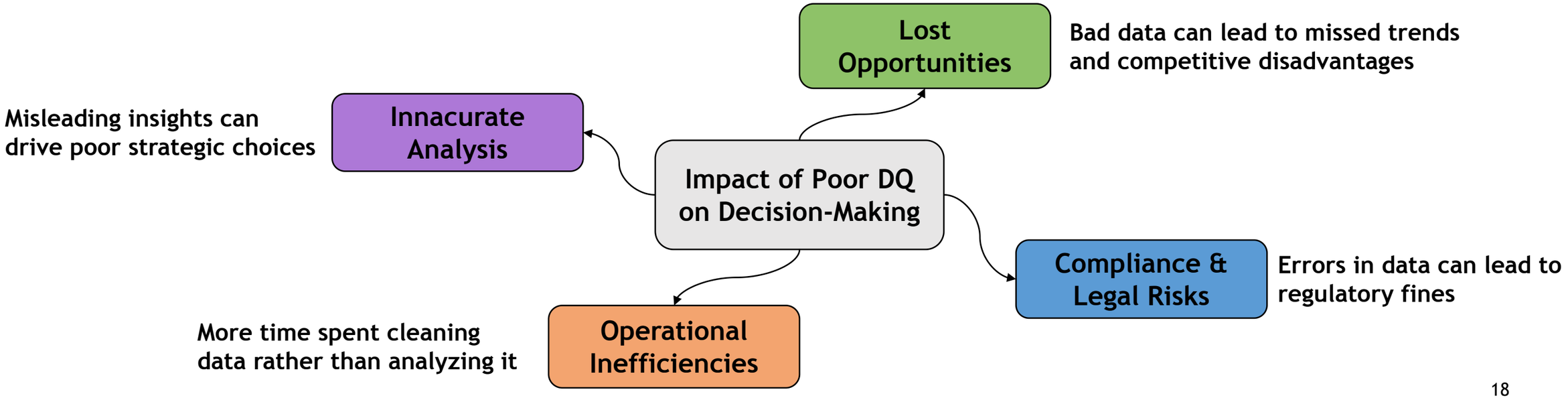


For LLMs and RAG systems:

- **Timeliness**: key for RAG to ensure up-to-date guidance
- **Consistency**: inconsistent terminology confuses models
- **Completeness**: missing context causes insufficient answers

Common Data Quality Issues and Business Impact

- Occur when data **does not meet quality requirements**
- Factors contributing to poor data quality:
  - **Source level issues:** Multi-source inconsistency, trustworthiness and unreliability
    - **Impact: Contradictory inputs leading to ambiguous recommendations**
  - **Generation level issues:** Human entry errors, sensor inaccuracies, missing values
    - **Impact: Garbage-in-garbage-out effect on machine learning models**
  - **Process level issues:** Collection and transmission failures
    - **Impact: Systematic errors that compound throughout the pipeline**



Data Wrangling

Data Complexity → **Data Wrangling** → Valuable Insights

- Data wrangling is the process of taking "raw" or "found" data and transforming it into something that can generate insight
- Raw data is often **incomplete, inconsistent, or unstructured**, requiring transformation before use
- Through wrangling, raw data is **cleaned**, **structured**, and **enhanced**

- **Without proper data quality assessment, wrangled data may still contain errors, leading to misleading insights.**

Data Cleaning

- Data is not ready to be processed as it is (raw form)
- Data should go through many stages, including pre-processing, to **ensure quality before further refinement and evaluation.**
- Addressing data quality early in the pre-processing stage is essential, especially in Big Data
- A key pre-processing activity (PPA) like **Data Cleansing** enhances data quality by identifying and correcting errors, handling missing values, and standardizing formats.
- The process of **Data Cleansing** typically includes the following steps:

01 Data Profiling

02 Handle Missing Data

03 Remove Duplicates

04 Standardize Formats

05 Fix Errors

06 Correct Inconsistent Data

07 Handle Outliers

08 Data Validation

09 Data Enrichment

⇒ **Data Cleaning is crucial for ensuring datasets are accurate, consistent, and ready for analysis.**

Data Quality Assessment Techniques

Definition : Systematic evaluation of data to determine if it meets the requirements of its intended use

Purpose: Ensure data is reliable, accurate, and fit for business, analytics, and decision-making

- Each assessment method vary in its effectiveness at evaluating different dimensions, such as:
  - Accuracy
  - Completeness
  - Consistency
  - Timeliness
  - Validity
  - Uniqueness
  - Integrity
- Maximum effectiveness is achieved when integrated combining multiple assessment approaches in a unified strategy

Assessment Techniques	Definition	Accuracy	Completeness	Consistency	Validity	Timeliness	Uniqueness	Integrity
Data Profiling	Examines structure, content, and quality and detect issues		x	x	x		x	x
Rule-Based Assessment	Verifies against predefined business rules	x		x	x			x
Data Matching	Identifies and merges duplicate or related records across datasets	x		x			x	
Data Scoring	Assigns numeric scores to dimensions	x						
Data Quality Audits	Analyzes data for compliance with business rules, policies, and standards	x	x	x		x		
Metadata Analysis	Evaluates the metadata definitions and consistency across systems		x	x	x			x
User Feedback Collection	Surveys of data consumers regarding perceived data quality/usability	x	x		x	x		

Data Profiling Technique

Definition:

- The process of analyzing data to understand its structure, content, patterns, and quality

Purpose:

- Provide insight into the condition and behavior of data
- Detect data quality issues such as duplicates, nulls, and inconsistencies
- Understand data patterns and relationships
- Act as a discovery or assessment phase before data processing or analysis

Outcome:

- Summary of the structure, content, and quality of a dataset
- Comprehensive view of your data's characteristics, including statistical metrics, patterns, distributions, and relationships

⇒ **Data Profile**

Data Profiling and Data Cleaning are distinct processes in the data management lifecycle but COMPLEMENTARY

Data Profiling Technique

- **“Health check”** or **“Fingerprint”** of your dataset that helps you understand what you're working with before deeper analysis or processing.
- A complete **data profile** typically includes:

Category	Metric / Element	Description	Example
Structure Information	Column Name	Name of the data field	`Customer_ID, Date_of_Birth`
	Data Type	Format type	`Integer, VARCHAR, Date`
	Data Length	Max string length or precision	`VARCHAR(50), Decimal(10,2)`
	Format Pattern	Expected format or pattern	`YYYY-MM-DD, +33 6 00 00 00 00`
Statistic Information	Count	Total number of rows	`10,000 rows`
	Min / Max	Lowest and highest values	`Min Age: 18, Max Age: 90`
	Mean / Median	Central tendencies	`Average Income: €32,000`
	Standard Deviation	Spread of the data	`Std Dev: 4.2`
	Percentiles	Distribution thresholds	`25th: 22, 50th: 35`
Completeness Metrics	Null Count	Number of missing (null) values	`2,350 nulls in Email`
	Fill Rate	Percentage of non-null entries	`76.5% filled`
	Empty Strings	Count of empty but non-null values	`" "`

Data Profiling Technique

Category	Metric / Element	Description	Example
Uniqueness Analysis	Unique Values	Count of distinct values	`9,000 distinct emails`
	Duplicates	Count of repeated values or rows	`1,000 duplicate records`
	Cardinality	Ratio of unique values to total records	`High cardinality in Invoice_ID`
Pattern Analysis	Pattern Detection	Format or structural consistency across values	`90% match phone pattern`
	Regex Validity	Conformance to specific regular expressions	`Email pattern match: 85%`
	Data Masking	Standard representations of data	`Date: DD/MM/YYYY`
Relationships Information	Primary Keys	Unique identifiers in the dataset	`Customer_ID`
	Foreign Keys	Cross-table references	`Order.Customer_ID → Customer.ID`
	Dependency Rules	Logical field dependencies	`If Age < 18 → No driver's license`
Quality Indicators	Domain Check	Values match allowed list or category	`Country in ['FR', 'BE', 'CH']`
	Temporal Logic	Time-based data validation rules	`Birth Date before Join Date`
	Frequency Tables	Most common values with percentages	`Gender: 60% F, 40% M`
	Histograms	Graphical summary of value distribution	`Age group bins`
	Outlier Detection	Identification of statistically rare or extreme values	`Age > 120 → flagged`

Practical Lab